Homework 1 (312)

93/100

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1.) $y' = f(x)g(y)$ $\quad \frac{dy}{dx} = f(x)g(y) \rightarrow \int \frac{dy}{g(y)} = \int f(x)\,dx \rightarrow G(y) = F(x) + C$

b) $y' + P(x)y = Q(x)$ $\quad \frac{dy}{dx} + P(x)y = Q(x)$ $\quad$ integrating factor: $\mu(x) = e^{\int P(x)dx}$

$$e^{\int P(x)dx}\frac{dy}{dx} + P(x)e^{\int P(x)dx}y = e^{\int P(x)dx}Q(x)$$

$$\int \frac{d}{dx}\left[e^{\int P(x)dx}y\right] = \int e^{\int P(x)dx}Q(x)$$

$$e^{\int P(x)dx}y = \int e^{\int P(x)dx}Q(x)\,dx + C$$

$$y = e^{-\int P(x)dx}\int e^{\int P(x)dx}Q(x) + Ce^{-\int P(x)dx}$$

C) $y'' + b^2y = 0$ $\quad$ let:

$$y = e^{mx},\ y' = me^{mx},\ y'' = m^2e^{mx}$$

$$m^2e^{mx} + b^2e^{mx} = 0$$

$$e^{mx}(m^2 + b^2) = 0$$

$e^{mx} \neq 0$, $m^2 + b^2 = 0$, $m = \pm bi$

$$y = Ae^{-bix} + Be^{bix}$$

using: $e^{i\phi} = \cos\theta + i\sin\theta$

$$y = A[\cos(-bx) + i\sin(-bx)] + B[\cos(bx) + i\sin(bx)]$$

$$= A\cos(bx) - A\sin bx + B\cos(bx) + Bi\sin(bx)$$

$$= (A+B)\cos(bx) + (B-A)i\sin(bx)$$

$$y = C_1\cos(bx) + C_2\sin(bx)$$

$C_1 = A + B$

$C_2 = (B - A)i$

D) $y'' - b^2y = 0$

$$r^2 - b^2 = 0$$

$$(r-b)(r+b) = 0$$

$x^2 - 25 = 0$

$(x-5)(x+5) = 0$

$x^2 + 5x - 5x - 25 = 0$

$x^2 - 25 = 0$

$$r = \pm b$$

$$y = C_1e^{-bx} + C_2e^{bx}$$

E) $y'' = b_0 + b_1x + b_2x^2$

$$\int \frac{d^2y}{dx^2} = \int (b_0 + b_1x + b_2x^2)\,dx$$

$$\int \frac{dy}{dx} = \int \left(b_0x + \frac{b_1x^2}{2} + \frac{b_2x^3}{3} + C_0\right)dx$$

$$y = \frac{b_0x^2}{2} + \frac{b_1x^3}{6} + \frac{b_2x^4}{12} + C_0x + C_1$$

F.) $\frac{1}{x}\frac{d}{dx}\left(x\frac{dy}{dx}\right) = b_0 + b_1 x + b_2 x^2$

$$\int d\left(x\frac{dy}{dx}\right) = \int b_0 x + b_1 x^2 + b_2 x^3 \, dx$$

$$x\frac{dy}{dx} = \frac{b_0 x^2}{2} + \frac{b_1 x^3}{3} + \frac{b_2 x^4}{4}$$

$$\int \frac{dy}{\cancel{dx}} = \int \left(\frac{b_0 x}{2} + \frac{b_1 x^2}{3} + \frac{b_2 x^3}{4} + \frac{C_0}{x}\right) dx$$

$$y = \frac{b_0 x^2}{4} + \frac{b_1 x^3}{9} + \frac{b_2 x^4}{16} + C_0 \ln x + C_1$$

G*) $\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) + b^2 y = 0$

* substitution

$y(x) = u(x)/x$

$$\frac{1}{x^2}\frac{d}{dx}\left(x^2 \frac{u(x)/x}{dx}\right) + b^2 \frac{u(x)}{x} = 0$$

quotient Rule

$$x^2 \frac{u(x)}{x\,dx} \Rightarrow \frac{x u'(x) - u(x)}{x^2}$$

left hand side of the plus sign

$$\frac{1}{x^2}\frac{d}{dx}\left(\cancel{x^2}\frac{x u'(x) - u(x)}{\cancel{x^2}}\right)$$

derivative

$$\frac{1}{x^2}\frac{d}{dx}\left(x u'(x) - u(x)\right)$$

product Rule

$$\frac{1}{x^2}\left(\cancel{u'(x)} + x u''(x) - \cancel{u'(x)}\right)$$

$$x\left[\frac{1}{x}\left(u''(x)\right) + b^2\frac{u(x)}{x} = 0\right.$$

$$u''(x) + b^2 u(x) = 0$$

making a guess of $u(x) = e^{rx}$

$$(r^2 + b) = 0$$

we get complex roots conjugate

$$r = \sqrt{-b} \quad i = \sqrt{b}$$

$$y = C_1 e^{\alpha x}\cos(\beta x) + C_2 e^{\alpha x}\sin(\beta x)$$

$$y = \frac{C_1\cos(\beta x)}{x} + \frac{C_2\sin(\beta x)}{x}$$

$\{\beta = b\}$

h*) $\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) - b^2 y = 0$

$$\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{(u(x)/x)}{dx}\right) - b^2\frac{u(x)}{x} = 0$$

Redoing the quotient Rule

$$\cancel{x^2} \cdot \frac{x u'(x) - u(x)}{\cancel{x^2}}$$

$$\frac{1}{x^2}\frac{d}{dx}\left(x u'(x) - u(x)\right)$$

Similar steps as g*

$$\frac{1}{x^2}\left(\cancel{u'(x)} + x u''(x) - \cancel{u'(x)}\right)$$

$$x\left[\frac{1}{x}\left(u''(x)\right) - b^2\frac{u(x)}{x} = 0\right.$$

$$u''(x) - b^2 u(x) = 0$$

guessing $u(x) = e^{rx}$

$$r^2 - b = 0$$

$$r = \pm b \quad i = \pm b$$

$$y = C_1 e^{\alpha x}\cos(\beta x) + C_2 e^{\alpha x}\sin(\beta x)$$

$$y = \frac{C_1}{x}e^{bx} + \frac{C_2}{x}e^{-bx}$$

$y'' + b_1 y' + b_2 y = 0$

$m^2 + b_1 m + b_2 = 0$

from the quadratic formula depending on the complex roots we get.

roots: $m_{1,2}: \alpha \pm i\beta \quad \beta > 0 \quad i^2 = -1$

Euler's Formula: $e^{i\theta} = \cos\theta + i\sin\theta$

$e^{i\beta x} = \cos\beta x + i\sin\beta x \qquad e^{-i\beta x} = \cos\beta x - i\sin\beta x$

$m_1 = \alpha \pm i\beta \quad m_2 = \alpha - i\beta$

$e^{i\beta x} + e^{-i\beta x} = 2\cos\beta x$

$e^{i\beta x} - e^{i\beta x} = 2i\sin\beta x$

cases (ii) $\Delta > 0$ & (iii) $\Delta = 0$ missing

(-3)

$y = b_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} \qquad C_1 = 1 \quad C_2 = -1$

$y_1 = e^{\alpha x}(e^{i\beta x} + e^{-i\beta x}) = 2e^{\alpha x}\cos\beta x$

$y_2 = e^{\alpha x}(e^{i\beta x} - e^{-i\beta x}) = 2ie^{\alpha x}\sin\beta x$

$y = C_1 e^{\alpha x}\cos\beta x + C_2 e^{\alpha x}\sin\beta x$

j) $y'' + 2xy' = 0$

sub. $\frac{dy}{dx} = u$.

power series form

$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$

$= a_n + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 +$

$\sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n + 2x\sum_{n=1}^{\infty} n a_n x^{n-1} = 0$

$\sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n + \sum_{n=1}^{\infty} 2n a_n x^n = 0$

$\sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n + \sum_{n=0}^{\infty} 2n a_n x^n = 0$

$a_0 + 2a_2 + \sum_{n=0}^{\infty} [(n+1)(n+2)a_{n+2} + (2n)a_n]x^n = 0$

$a_0 + 2a_2 = 0 \quad (n+1)(n+2)a_{n+2} + 2n a_n = 0$

$\Rightarrow a_{n+2} = \frac{-(2n)a_n}{(n+1)(n+2)}$

@ $n=0$ $a_{0+2} = \frac{-(2\cdot 0)a_0}{(1)(2)} = 0 \quad a_2 = 0$

@ $n=1$ $a_{1+2} = \frac{-(2\cdot 1)a_1}{(2)(3)} = \frac{-2a_1}{6} = a_3$

@ $n=2$ $a_{2+2} = \frac{-(2\cdot 2)a_2}{(3)(4)} = \frac{-4a_2}{12} = a_4 = 0$

@ $n=3$ $a_5 = \frac{-2(3)a_3}{(3+1)(3+2)} = \frac{-6a_3}{(4)(5)} = \frac{-6a_3}{20} = a_5$

@ $n=4$ $a_{4+2} = \frac{-2(4)a_4}{(4+1)(4+2)} = \frac{-8a_4}{30} = a_6 = 0$

$(5)(6)$

@ $n=5$ $a_{5+2} = \frac{-2(5)a_5}{(5+1)(5+2)} = \frac{-10a_5}{(6)(7)} = \frac{-10a_5}{42} = a_7$

$\therefore a_8 = 0$

$\therefore a_9 = \frac{-14}{(9)(8)} \cdot \frac{-10}{(7)(6)} \cdot \frac{-6}{(5)(4)} \cdot \frac{-2}{(3)(2)} a_1$

$y = \sum_{n=0}^{\infty} a_n x^n$

$y = a_0 + a_1 x + 0 + \left(\frac{-2}{3\cdot 2}\right)a_1 x^3 + 0 + \left(\frac{-6}{(5)(4)}\right)\left(\frac{-2}{3\cdot 2}\right)a_1 x^5 + 0$

$\cdots$

$y = a_0 + a_1 x + \sum_{n=1}^{\infty} \frac{-2\cdot -6\cdot 10\cdot (4m-3)}{(2m+1)!} \cdot x^{2m+1}$

Incorrect solution

(-4)

K.) $\frac{d}{dx}\left[f(y)\frac{dy}{dx}\right]=0$

$$\int d\left[f(y)\frac{dy}{dx}\right]=\int 0\,dx$$

$$\int d\, f(y)\frac{dy}{dx}=\int C_0\cdot dx$$

$$\int f(y)\,dy = C_0x+C_1$$

L.) $x^2\left[\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right)=b_0+b_1x+b_2x^2\right.$

$$dx\left[\frac{d}{dx}\left(x^2\frac{dy}{dx}\right)=b_0x^2+b_1x^3+b_2x^4\right.$$

$$\int d\left(x^2\frac{dy}{dx}\right)\int(b_0x^2+b_1x^3+b_2x^4)\,dx$$

$$\frac{x^2\frac{dy}{dx}=\frac{b_0x^3}{3}+\frac{b_1x^4}{4}+\frac{b_2x^5}{5}+C_0}{x^2}$$

$$\int\frac{dy}{dx}=\int\left(\frac{b_0x}{3}+\frac{b_1x^2}{4}+\frac{b_2x^3}{5}+\frac{C_0}{x^2}\right)dx$$

$C_0x^{-2}\rightarrow C_0x^{-1}$

$\frac{-C_0}{x^2}$

$$y=\frac{b_0x^2}{6}+\frac{b_1x^3}{12}+\frac{b_2x^4}{20}-\frac{C_0}{x}+C_1$$

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